Advanced Quantum Mechanics

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Summer 2026
Exercise sheet 1

Problem 1.1: Energy measurement

Consider a quantum system described by the Hamiltonian

$$H = \hbar\omega \begin{pmatrix} \frac{17}{9} & \frac{2\sqrt{2}}{9}e^{-2i} & 0 & 0 \\ \frac{2\sqrt{2}}{9}e^{2i} & \frac{10}{9} & 0 & 0 \\ 0 & 0 & \frac{7}{2} & \frac{1}{2}\,e^{i\pi/3} \\ 0 & 0 & \frac{1}{2}\,e^{-i\pi/3} & \frac{7}{2} \end{pmatrix} . \tag{1}$$

The system is in the state

$$|\psi\rangle = \frac{1}{2}\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} . \tag{2}$$

a) What are the possible outcomes of measuring the energy?
Hint: Note that the matrix is block-diagonal.

b) What are the probabilities for each of these outcomes?

Problem 1.2: Spins in a magnetic field

A beam of electrons enters a homogeneous magnetic field with strength $B = |\vec{B}|$ at time $t = 0$. The magnetic field is pointing in z-direction. The corresponding Hamiltonian is,

$$H = \frac{2\mu_B}{\hbar} B S_3 \,, \tag{3}$$

where $\vec{S}$ is the spin operator and $\mu_B = \frac{e\hbar}{2m_e}$ is the Bohr magneton. You can assume that these are matrices in the conventional basis (z- or S_3-basis),

$$|\!\uparrow\rangle \equiv \begin{pmatrix} 1 \\ 0 \end{pmatrix} , \qquad |\!\downarrow\rangle \equiv \begin{pmatrix} 0 \\ 1 \end{pmatrix} . \tag{4}$$

$|\!\uparrow\rangle$ ($|\!\downarrow\rangle$) means that the spin points in positive (negative) z-direction. In this case, the spin operator can be written in terms of the conventional Pauli-matrices, $\vec{S} = \frac{\hbar}{2}\vec{\sigma}$.

At the initial time $t = 0$, the electrons are polarized in positive x-directions, i.e. they are eigenstates of S_1 with eigenvalue $\frac{\hbar}{2}$.

a) What is the state of the electrons at a time $t > 0$?

b) What is the expectation value of the spin in x-direction, $\langle S_1 \rangle$, at time t?

c) What is the probability of measuring the value $-\frac{\hbar}{2}$ if the spin in y-direction, S_2, is measured at time t?

Problem 1.3: Harmonic oscillator

Consider a system in the following state

$$|\psi\rangle = \frac{|n\rangle + |m\rangle}{\sqrt{2}}\,, \tag{5}$$

where $n, m \in \mathbb{N}$ and $|n\rangle$ is the n-th eigenstate of the harmonic oscillator.

a) Compute the expectation values of the position and momentum operators, $\langle X\rangle$, $\langle P\rangle$, and their squares, $\langle X^2\rangle$, $\langle P^2\rangle$.
Hint: Express X and P in terms of ladder operators.

b) Show that the uncertainty relation is fulfilled for all n, m.